Product Requirements Document:

Version: 2.0

1. Introduction & Vision

1.1. Vision Statement:

To create a future where design and product teams can achieve the depth and reliability of moderated usability testing without being limited by scheduling constraints, time inefficiencies, or high research costs, through the power of AI. Userlenz aims to make high-quality research faster, more accessible, and consistently actionable, enabling teams to make confident, user-centered decisions at any stage of product development.

1.2. Problem Statement:

User research is essential for building successful products, yet in practice, it is often perceived as slow, costly, and unreliable in delivering results. Conducting moderated usability testing often requires significant time to coordinate schedules across teams and participants, and even small delays or mismatched availability can push research timelines well beyond plan. As a result, research that should be completed quickly often stretches into days or weeks.

Beyond scheduling, teams also spend substantial time preparing studies and analyzing outcomes. Defining the right tasks, writing unbiased questions, and structuring effective test flows require experience and effort that many teams lack. After testing, qualitative analysis, such as reviewing recordings, interpreting user behavior, and synthesizing findings into actionable insights, becomes another major time investment.

From a cost perspective, high-quality research often depends on hiring professional moderators, researchers, or external agencies, making it difficult for smaller teams or fast-moving products to run research consistently. When unmoderated testing is used as a cheaper alternative, the results are often perceived as lower quality due to limited visibility into user behavior, expressions, and decision-making context.

These challenges lead to a critical outcome: research is delayed, deprioritized, or trusted less than it should be. Teams either skip research altogether or proceed with incomplete or low-confidence insights, even though faster, more scalable approaches should, in theory, make research more accessible. The core problem lies not in the value of research itself, but in the trade-off among time, cost, and result quality.

1.3. Solution Overview:

Userlenz is an AI-powered user research platform designed to address the core challenges of time, cost, and quality of results. It reduces the time required to plan, run, and analyze studies by automating manual research tasks that traditionally slow teams down, from study preparation to insight synthesis.

By minimizing scheduling dependencies and reducing the need for professional moderators, Userlenz lowers research costs while enabling teams to run studies more frequently. At the same time, the platform is designed to help unmoderated research deliver depth and reliability comparable to moderated studies, allowing teams to generate high-quality, decision-ready insights without the operational overhead typically associated with traditional research methods.

1.4. Target Audience:

Designers, User Researchers, Product Managers, Startup Founders, Marketers.

2. Core Features & User Flow

2.1. Feature 1: AI Assisted Study Builder (MVP)

- **User Problem:** "I know I need to test my new feature, but I'm not sure what questions to ask or which research method is best. Writing a good study plan takes hours."
- **Description:** A guided module that helps users create effective research studies (surveys, usability tests, interview scripts) from a simple goal. The user inputs their research objective in natural language (e.g., "I want to understand why users are dropping off during our checkout process").
- **AI Integration & Manual Work Reduction:**
  - **Method Recommendation:** The AI analyzes the user's goal and suggests the most appropriate research method (e.g., "A 5-user usability test would be effective for identifying friction points").
  - **Question Generation:** The AI generates draft, relevant, and unbiased questions and task prompts from a library of UX best practices and the provided context. For a checkout flow, it might generate tasks such as "Add a product to your cart and proceed to payment" and follow-up questions such as "Was there any point where you felt unsure about what to do next?"
  - **This reduces:** The manual work of brainstorming questions, researching best practices, and structuring study scripts from scratch.
- **User Flow AI-Assisted Study Builder:**
  1. **Entry Point: Home – AI Chat Input:** A conversational input that allows users to describe their research needs using natural language.

     User Action:
     - Users land on the Home screen.
     - Users type a research request in the AI chat input, for example: “I want to test the shopping experience of my fashion mobile app prototype, especially navigation, product discovery, and checkout.”

     System Behavior (As the user types):
     - Prompt suggestions appear below the input, similar to any AI suggestions, such as:

        - Test usability of a mobile checkout flow
        - Evaluate product discovery experience.
    - The system detects user intent and research scope in real time.

2. **Study Type Selection**

    User Action:
    - User selects a study type from available templates, such as:
        - Usability Testing
        - Satisfaction Survey
        - Feedback Survey
    - Users can enhance or refine the prompt before proceeding.
    - Users click "Generate" to continue.

    System Behavior:
    - The AI chat prompt updates to reflect the chosen study structure.

3. **AI Processing & Goal Generation**

    System Behavior after the user submits the prompt:
    - AI analyzes the user's research needs and selected study type.
    - AI generates multiple possible research goals, each representing a different research focus, for example:
        - Identify navigation friction in the shopping flow
        - Understand product discovery behavior
        - Detect drop-off points during checkout

4. **Goal Review & Canvas Preview**

    Review Your Goals Panel (Right Side):
    - Generated goals appear in a "Review your goals" panel on the right side of the screen.
    - Each goal represents a complete study direction.

    Canvas Preview Behavior (Left Side):
    - Users click on a goal in the right panel

    System Behavior:
    - The canvas on the left updates instantly to reflect the selected goal and acts as a live preview
    - The canvas displays a fully generated study flow, consisting of structured research blocks.
    - Each goal selection results in a different canvas structure and content, allowing users to visually compare study approaches.

    User Capabilities:
    - Users can:

        - Switch between different goals
        - Preview how each goal translates into a complete study flow on the canvas

5. **Goal Selection**

   Constraint:
   - Users can select only one goal to proceed.

   User Action:
   - Users click on a goal in the right panel

   System Behavior:
   - The selected goal becomes the active study.
   - The generated blocks remain on the canvas as the editable study structure.
   - Other generated goal variations are discarded.

6. **Regenerate Study Flow**

   User Action:
   - User opens the “More” (···) menu next to the study name in the top bar and selects “Regenerate”.

   System Behavior:
   - The system regenerates multiple goal variations, similar to the initial goal generation step.
   - The currently selected goal is preserved and labeled as “Your current goal”.
   - Newly generated goals are displayed alongside the current goal in the “Review your goals” panel.

   User Capabilities:
   - Users can:
     - Compare the current goal with newly generated alternatives
     - Preview each goal as a full study flow on the canvas
     - Select one goal to proceed

   Goal Selection Constraint:
   - Only one goal can be active at a time.
   - When a new goal is selected:
     - The canvas updates to the newly chosen study flow
     - The previously active goal is replaced

2.2. Feature 2: AI Moderated Engine - Preview Study (MVP)

- **User Problem:** "I want to run unmoderated research to save time and cost, but the

results often feel shallow. Without a moderator, I can't observe user behavior, ask follow-up questions, or understand why users act the way they do."

- **Description:** The AI Moderated Engine enhances unmoderated research by introducing AI-delivered voice instructions during preview and live testing. Participants are guided through each task using clear, structured voice instructions, creating a consistent research experience without requiring live human moderation. During the session, participant actions and interactions are recorded. These recordings then become structured inputs for AI-driven analysis, enabling a deeper understanding of user behavior beyond surface-level responses.
- **AI Integration & Manual Work Reduction:**
  - **Voice-Based Task Instructions:** The AI delivers consistent, clear voice instructions that explain tasks and expectations, reducing the need for live facilitation.
  - **Standardized Study Flow:** The AI controls task sequencing and pacing via voice prompts, ensuring all participants follow the same study flow.
  - **Behavior Recording for AI Analysis:** User interactions and session recordings are captured per participant and later analyzed by AI to identify patterns, friction points, and behavioral signals.
  - **Reduced Interpretation Effort:** By feeding behavioral recordings directly into AI analysis, researchers spend less time manually reviewing sessions and synthesizing findings.

- **User Flow AI Moderated Engine:**

  The AI Moderated Engine in the Preview Study allows researchers to experience the study exactly as participants will, **while retaining researcher-only controls**. Unlike participants, researchers can **freely jump between blocks** to validate flow, content, and interactions without following a strict sequence.

1. **Preview Only Navigation (Key Difference from Participant Flow):** Unlike participants, researchers are **not required to complete tasks sequentially**.

   User Actions:

   - Jump to any block in the study
   - Restart preview from the beginning
   - Exit preview and return to Canvas

   System Behavior:

   - Jumping between blocks:
     - Does not require completing previous tasks
     - Does not affect study logic or data
   - Preview navigation controls are always visible.
   - Participant experience remains strictly sequential in live studies.

2. **Entry Point - Canvas "Preview Study":** The preview is launched from the Canvas and initializes the AI Moderated Engine in Preview Mode, allowing

researchers to experience the study without recording any data.

User Actions:

- User is on the **Canvas** with a completed study flow.
- User clicks "**Preview Testing**".
- Preview opens in AI Moderated Mode, simulating the participant experience with AI moderation enabled.

System Behavior:

- The system loads the study in **Preview Mode.**
- AI Moderated Engine is activated.
- The experience mirrors the participant flow, with additional preview-only navigation controls.
- **No data is recorded**.

3. **Welcome & Language Selection Screen (Welcome Screen):** The preview begins with the same **combined Welcome + Language Selection screen** used in the participant experience.

   User Actions:

   - User selects preferred language
   - User clicks "**Get Started**".

   System Behavior:

   - AI moderator introduces the study:
     - Explains the purpose
     - What participants will do
     - Estimated time (if possible)
   - AI speech adapts to the selected language
   - Spoken introduction is displayed as synchronized transcript text.

4. **Task Block Introduction (Per Block):** For each block in the study flow, the AI moderator guides the experience.

   User Actions:

   - User views the task block screen.
   - User waits or initiates playback if manual play is enabled.

   System Behavior:

   - The AI moderator:
     - Reads the **task title and task description** aloud.
     - Uses neutral, non-leading language.
   - Audio includes:
     - Play/pause controls
     - Waveform visualization
   - The **full transcript** of the AI-read content is displayed on screen and

synchronized with audio.

5. **Task Block Experience (Interaction Enablement):** After the AI finishes reading the task instructions, the interaction area becomes active based on the block type.

   User Actions:

   - User listens to the AI moderator.
   - User prepares to start the task.
   - User reads the transcript if needed.
   - User prepares to start the task.

   System Behavior:

   - Once the AI finishes reading:
     - The **interaction area becomes active**.
     - The interaction type is loaded based on the block configuration (e.g., Prototype, Question input, Rating or scale).

6. **Task Block Experience (During Interaction):** Researchers interact with the task exactly as participants would, allowing them to validate the interaction flow without capturing any data.

   User Actions:

   - User interacts with the prototype or task content exactly as a participant would.
   - User completes the required interaction.

   System Behavior:

   - System mirrors participant interaction rules.
   - **No interaction data is saved** in Preview Mode.
   - AI remains silent unless follow-up guidance is defined in the block.

7. **Task Block Experience (Task Completion & Transition):** Once a task is completed, the AI acknowledges completion and guides the transition to the next task or selected block.

   User Actions:

   - User completes the task.
   - User clicks **Next / Continue**.
   - User can choose to **jump to another block** using the preview controls.

   System Behavior:

   - If continuing sequentially:
     - AI verbally acknowledges task completion.
     - AI introduces the next task.
     - Transcript updates accordingly.

        - Progress indicator advances.
    - If jumping to another block:
        - The system immediately loads the selected block.
        - AI introduces the newly selected task from the beginning.
        - Progress indicator updates to reflect the selected block position.

8. **Completion & Wrap-Up (Thank You Screen)**

   System Behavior:

   - AI moderator delivers a closing message verbally.
   - The closing transcript is displayed.
   - Confirms preview completion.
   - User can exit the preview and return to Canvas.

2.3. Feature 3: AI Moderated Engine - (MVP POV Participants)

- **Description:** The AI Moderated Experience provides participants with a guided testing flow, delivered via AI-generated voice instructions, during preview and live research sessions. Instead of reading long text instructions or navigating unclear tasks, participants receive clear, step-by-step guidance that helps them understand what to do at each stage of the study.
- **AI Integration & Participant Experience:**
    - **Voice-Guided Instructions:** Participants receive clear voice instructions that explain tasks, expectations, and transitions throughout the session.
    - **Predictable and Structured Flow:** The AI ensures a consistent sequence of tasks, reducing confusion and helping participants stay focused during the study.
    - **Reduced Cognitive Load:** By relying on voice guidance rather than dense text instructions, participants can focus more on completing tasks naturally.
    - **Behavioral Recording Awareness:** Participants complete the study knowing their interactions are being recorded for research purposes, increasing transparency and compliance.
- **User Flow AI Moderated Engine - Participant Experience:**

  The Participant Testing Preview simulates the live participant experience exactly as it will occur during an active study. Unlike the researcher preview, participants **must complete all tasks in a fixed sequence and cannot skip or jump between blocks**. This ensures consistency, data integrity, and controlled exposure to tasks.

  1. **Entry Point - Participant opens the study link:** Participants access the study via a shared link provided by the researcher. This link opens the AI Moderated Engine directly in Participant Mode.

     Participants Actions:

     - User clicks the shared study link
     - Study opens in the browser (or in-app web view)

     System Behavior:

- System validates the study link
- Loads the study in Participant Mode
- Initializes the AI Moderated Engine
- Applies participant-specific constraints:
    - Sequential task flow enforced
    - No preview-only controls
- Begins at the Welcome Screen

2. **Welcome & Consent Screen:** The participant experience begins with a combined Welcome and Language Selection screen.

   System Behavior:

   - AI moderator introduces the study via voice:
     - Briefly explains the purpose of the study
     - What the participant will be asked to do
     - Approximate session duration
   - Recording notice is clearly communicated:
     - Screen interactions and responses will be recorded
   - Spoken content is displayed as synchronized transcript on screen.

   Participant Actions:

   - Participant selects preferred language (if applicable).
   - Participant confirms and clicks "**Get Started"**.

3. **Task-by-Task Guided Experience:** Participants must complete tasks in the predefined order without skipping.

- **Task Introduction**
  System Behavior:

  - AI moderator:
    - Reads the task title and instructions aloud
    - Uses neutral, non-leading language
  - Audio UI includes:
    - Play/pause controls
    - Waveform visualization
  - Full transcript of the spoken instructions is visible on screen.

  Participant Actions:

  - Participant listens to the instructions
  - No interaction is enabled during instruction playback

- **Interaction Phase**
  System Behavior:

  - Once the AI finishes speaking:
    - The interaction area becomes active

        - The relevant interaction task is loaded.

    Participant Actions:

    - Participant completes the task naturally
    - Interacts with the prototype or answers questions as instructed

  - **Task Completion & Transition**
    System Behavior:

    - AI verbally acknowledges task completion
    - Introduces the next task using voice guidance
    - Transcript updates in real time
    - Progress indicator advances
    - System transitions smoothly to the next block

4. **Completion & Wrap-Up (Thank You Screen)**

    System Behavior:

    - AI moderator delivers a closing message:
        - Thanks to the participant for their time
        - Confirms the session is complete
    - The final spoken message is transcribed
    - System finalizes and records all participant data
    - Participant's session is marked as completed
    - After the Thank You message is shown, the system displays a signup form.

2.4. Feature 4: AI Synthesis/Analysis Engine (MVP)

- **User Problem:** "After running research sessions, teams still spend significant time reviewing recordings, observing user behavior, and synthesizing findings, making the process time-consuming and difficult to manage."
- **Description:** The AI Synthesis / Analysis Engine analyzes research data after a study has been completed and shared with participants. It processes both research outcomes and recorded user behavior during testing, helping teams understand not only what users did or said, but how they interacted throughout the session. By analyzing behavioral recordings alongside research results, the AI supports the synthesis work typically performed during moderated research, transforming raw sessions into structured, meaningful insights.
- **AI Integration & Manual Work Reduction:**
  - **Research Outcome Analysis:** Responses, task results, and feedback collected during the study are analyzed together with behavioral data to provide contextual understanding.
  - **Cross-Session Pattern Detection:** The AI aggregates data across participants to surface recurring behaviors, common issues, and shared responses.
  - **Insight Structuring:** Findings are organized into clear, digestible outputs that help teams quickly understand what happened during the research and why it

matters.

2.5. General User Flow: From Goal to Report (MVP)

1. **Sign Up / Login:** Users create an account or log in.
2. **Onboarding:** Users answer brief onboarding questions and get a quick intro to how Userlenz works.
3. **Dashboard:** Users land on the dashboard where they can describe their research needs using AI, choose templates, or open recent studies.
4. **Select Goal (AI Recommendations):** Based on the user's input, AI suggests the top 3 research goals. Users review, pick one, or regenerate new options.
5. **AI Generates Study on Canvas:** AI builds the study automatically with tasks and questions. Users can edit, add, remove, or reorder anything.
6. **Preview Study:** Users preview the study experience to make sure everything is clear and works as expected before starting to launch it.
7. **Launch Study:** Users either share a link with their own participants or hire participants directly through Userlenz, and can monitor participation.
8. **Participant Testing:** The AI Moderated Experience provides participants with a guided testing flow, delivered via AI-generated voice instructions, during preview and live research sessions.
9. **Results (Real-Time Data):** Responses update in real time. Users can view performance metrics, manage participants, and refine collected data.
10. **AI Insight Report:** Once complete, the AI automatically generates a polished insight report with summaries, themes, visuals, and shareable outputs.

3. MVP Flow General Details

3.1. Register flow

- **Sign Up**
  - User signs up via email, Google, or workspace SSO.
  - Forget password.
- **Onboarding**

  Basic onboarding questions:
  - Full name, company name, and size.
  - Role (Designer / PM / UX Researcher / UX Copy Writer / Other)
  - Primary purpose (Usability Testing / Concept Validation / User Interviews / etc.)

Onboarding answers will **personalize the user's dashboard** experience (Study Template Recommendation).

3.2. Dashboard (New User)

- **New User Onboarding**

  A quick intro to how Userlenz works

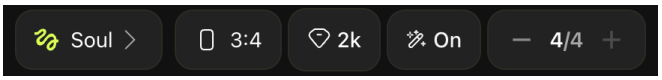
- **Sidebar Menu**

  - Home
  - Project
  - Study Template Library (Need to add and define the available templates)
  - Credits/Plan (remaining credits & bisa untuk beli credits)
  - Explore plan (see all plans or upgrade plan)
  - Profile & Settings
  - Help center (mostly articles, minimalization of user sending tickets)

- **AI Chat Input**

  Describe the research need in natural language

  - User **explains requirement** in AI chat (e.g., I want to test the shopping experience of my fashion mobile app prototype, especially navigation, product discovery, and checkout.)
  - User can **select a study type** based on available templates (Usability Testing, Satisfaction Survey, Feedback Survey, etc.)
  - When the user types a prompt, **suggestion prompts** will appear below it.
    ex: chatgpt or any AI
  - The **input fields dynamically update** based on study type.

    ex: https://higgsfield.ai/image/soul
  - User can **enhance** the prompt if needed.
  - User can **upload a document or file**

- **Userlenz Study Templates**

  Based on your onboarding answers, we'll show a personalized list of templates beneath the AI chat.

  - **Usability Testing:**
    - Usability testing a new product
    - Run an early prototype test
    - Feature discoverability template
    - Collect insights on features
    - Test feature usability
    - Test mobile app usability
    - Validate live website sign-up funnel flow
    - Test live website navigation

- Test your website sign-up flow
- Run a Single Ease Question (SEQ) test
- System Usability Scale (SUS) Template

available UT template from the maze.

ex design: https://app.heygen.com/home

3.3. Dashboard (Existing User)

- **Sidebar Menu**
  - Home
  - Project
  - Study Template Library (Need to add and define the available templates)
  - Credits/Plan (bisa untuk beli credits or change plan)
  - Widget for AI Credits/Plan (remaining credits/plan date expired)
  - Profile & Settings
  - Help center (mostly articles, minimalization of user sending tickets)

- **AI Chat Input**

  Describe the research need in natural language

  - User **explains requirement** in AI chat (e.g., I want to test the shopping experience of my fashion mobile app prototype, especially navigation, product discovery, and checkout.)
  - User can **select a study type** based on available templates (Usability Testing, Satisfaction Survey, Feedback Survey, etc.)
  - When the user types a prompt, **suggestion prompts** will appear below it.
    ex: chatgpt or any AI
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    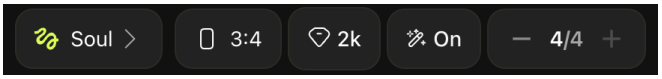

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  - User can **upload a document or file**

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    - Feature discoverability template
    - Collect insights on features

- Test feature usability
- Test mobile app usability
- Validate live website sign-up funnel flow
- Test live website navigation
- Test your website sign-up flow
- Run a Single Ease Question (SEQ) test
- System Usability Scale (SUS) Template

available UT template from the maze.

ex design: https://app.heygen.com/home

- **Recent Studies (Dynamic)**

  Displays the existing user's most recent research projects in a quick-access list, allowing them to seamlessly continue where they left off.

3.4. Select Goals

AI suggests the **Top 3 Research Goals** based on user input.

- **Each goal includes:**
  - Description of objective
  - Preview of test flow
- **User can:**
  - Select one goal
  - View the detailed flow of each goal
  - Or **Regenerate 3 New Goals**

3.5. Assign Study to Project

Before generating the study, UserLenz asks the user to:

- **Select an existing project,** or
- **Create a new project**
  - Input project name

Ensures the studies are always organized within a project context.

3.6. AI Generates Study on Canvas

After goal selection, AI automatically builds a structured flow on the **Canvas.**

- 📌 **Project Controls**

Top bar includes:

- Navigation controls:
  - Version history (history goals and block)
  - Device settings (Request response on Desktop, Tablet, Mobile)
  - Voice settings?
  - Edit (Shortcut Undo, Redo, Copy, Paste, Find content, select all, lock)
  - View (Shortcut zoom,
  - Arrange (Align shortcut)
  - Help center

- AI Credit Widget (remaining credits for AI Gen & AI Voice)
- Project name
- Study name the user can:
  - Rename the study name
  - View goals description and regenerate goals (the block's detail change)

- 🚀 **Study Controls**
  At the top right:
  - Online users on the project, we can share the canvas link and give permission to edit or view, like Figma (after MVP)
  - Preview study
  - Start testing or launch

- 🧩**Visual Flow Builder**
  The Visual Flow Builder is a **node-based canvas** that visually represents the full participant journey, allowing users to design, understand, and refine studies with clarity and control.

  Canvas Overview:
  - The canvas sits at the center of the workspace.
  - The study is represented as a linear, **node-based** flow.
  - Each **block = one participant step/task**.
  - **Welcome and Thank You blocks** are included by **default** and cannot be removed.
  - Blocks are connected sequentially, showing the exact order participants will experience.

  Block Structure:
  Each block contains:
  - **Block Title** (e.g., Task 1, Question, Scenario)
  - Prompt input area: Users describe what the participant should do or answer.
  - AI Actions:
    - **Generate** → AI creates task content.
    - **Enhance** → AI improves overall block inputs.

    - Each block supports a conditional state depending on the user's action.
    - Block Controls:
        - Duplicate (Copy/Paste)
        - Delete
        - Change block type
    - "+" buttons allow users to insert a new block between existing ones, keeping editing flexible and intuitive.

- 🎛️**Right-Side Editor Panel**
    - **When a node is selected**, a detailed editor appears on the right panel:
        - Allows editing the input, such as **prompt, title, description, and question or task content.**
        - Includes **Form Editor** and **Preview modes**, enabling users to easily toggle between editing the study and seeing how it will appear to participants.
    - **When the comment icon** on the bottom controls is clicked appears on the right panel:
        - Showing comments list
        - Search comments
        - Filter comments

- 🧭 **Toolbar**
  Bottom controls include:
    - Move (Cursor arrow & hand)
    - Comments
    - Add block
    - Zoom level

3.7. Preview Study

This allows researchers to fully experience the study exactly as participants will, ensuring the flow, content, and interactions are accurate, accessible, and ready before launch.

- **Language Selection (Welcome Screen – Entry Step)**

  The preview begins with the same combined **Welcome + Language Selection screen** shown to participants.

  Structure:
    - Study overview, participation instructions
    - Language selector
    - Primary CTA "Get Started"

- **Audio Instruction Playback (Preview Mode)**

  Researchers can preview all audio instructions used in the study.

Includes:
  - Play/pause controls
  - Waveform visualization

- **Transcript Support**

  Displays written transcripts alongside audio to ensure accessibility and consistency.

  Enables researchers to:
  - Check accessibility
  - Verify translation accuracy
  - Ensure consistency between audio and text

- **Sequential Block Validation**

  Shows each study block in its real order to confirm narrative flow and task progression.

- **Interaction Simulation**

  Researchers interact with prototypes or any interaction exactly as participants would.

- **Progress Awareness**

  Progress indicator mirrors participant experience.

- **Preview Controls**

  Preview-only navigation tools:
  - Jump to specific block
  - Restart preview
  - Exit preview back to Canvas

3.8. Launch Study (Modal)

The **Launch Study modal** allows users to start, control, and monitor a study from a single centralized interface. It is accessed from the **Study Control** area via the **"Launch"** button located at the top-right of the study view.

- **Access:**
  - Entry point: **Study Control → Launch button**
  - Trigger: Clicking **"Launch"** opens a modal overlay
  - Modal is accessible both before and after the study starts

- **Launch Modal Structure**
  - Shareable testing link section
  - Hire Participant (future feature)
  - Primary CTA
    - "**Start testing**" button (default).
    - Changes to "**Stop testing**" once the study is live.

- Performance Table (Real-Time)
    - Monitor the performance table in real-time
    - Study status (live, stopped)

3.9. Participant Testing Experience

This experience guides participants through the study in a clear, accessible, and distraction-free flow, ensuring they understand expectations and can participate confidently.

- **Language Selection (Welcome Screen – Entry Step)**

  The first screen participants see combines the study **introduction and language preference selection.**

  Structure:
  - Study overview, participation instructions
  - Language selector
  - Primary CTA "Get Started"

- **Audio Instruction Playback (Preview Mode)**

  Audio instructions are presented at the beginning of relevant tasks.

  Includes:
  - Play/pause controls
  - Waveform visualization

- **Transcript Support**

  Displays written transcripts alongside audio to ensure accessibility and consistency.

- **Sequential Block Validation**

  Shows each study block in its real order to confirm narrative flow and task progression.

- **Interaction Simulation**

  Participants interact with prototypes or engage in direct interaction.

- **Progress Awareness**

  A subtle progress indicator that shows completion status, helps participants manage time, and reduces uncertainty.

- **Thank You & Completion Screen**

  Confirms successful completion.

3.10. Result (Real-Time Data)

The Results section is the "Command Center" where raw participant data is transformed into a high-fidelity map of user behavior. While Maze provides static metrics, Userlenz provides **Contextual Intelligence** where the AI does the manual work for you.

A. General Methods (Included in EVERY study)

These metrics provide a high-level overview of how the study is performing overall. We go beyond basic counts to provide a health check on your data quality.

- **Advanced Participant Tracking:**
  - **Live Counter:** Total participants (Completed, In-Progress, Screened Out).
  - **Health Score (0-100):** AI evaluates each participant's reliability. It flags "Speeders" (who click too fast) or "Low Effort" responses (who type "good" or "nice" in text boxes).
  - **Demographic Snapshot:** Instant breakdown of device type, browser, and geographic location.
  - *Example:* "Participant #24 flagged for 'Speeding'—completed a 5-minute task in 40 seconds. Data excluded from success average."
- **Global Completion & Funnel Analysis:**
  - **Drop-off Thermal Map:** A visual funnel showing exactly at which block participants are quitting.
  - **Average Time on Study:** Real-time calculation of how long the average session takes vs. your estimated time.
  - *Example:* "Significant drop-off (60%) detected at Block 4. Users are likely finding the 'Legal Consent' document too long."
- **Sentiment & Text Synthesis:**
  - **Emotional Pulse:** AI reads all text inputs to categorize the "Mood" (Frustrated, Satisfied, Confused).
  - **Keyword Cloud:** Automatically identifies the most common words used in open-ended answers.
  - *Example:* "The word 'Confusing' appeared in 45% of responses related to the checkout button."

B. Specific Methods (Deep-Dive for Usability Testing)

These are advanced tools used specifically when testing prototypes or live websites. We provide deeper behavioral layers than standard tools.

- **Friction & Behavior Scoring:**
  - **Rage Click Detection:** Automatically flags areas where users clicked rapidly in frustration.
  - **Hesitation Score:** Measures the time between a page loading and the first action. High hesitation indicates a lack of "Information Scent."
  - **Interaction Velocity:** AI measures the speed of mouse movement. Erratic movement suggests a "Lost" user.
  - *Example:* "Task 2 has a High Friction Score (78/100) due to 15 seconds of hesitation before the first click."

- **Advanced Interaction Mapping:**
    - **Standard Click Map:** Visual frequency of all clicks.
    - **Confusion Map (AI Exclusive):** Highlights clicks on elements that are **not** interactive (e.g., clicking on a decorative icon thinking it's a link).
    - **Scroll & Attention Maps:** Shows how far users scrolled and where their cursor "hovered" the most, indicating visual attention.
    - *Example:* "Confusion Map shows 40% of users tried to click the 'Free Shipping' banner, which is currently non-interactive."
- **Smart Path Analysis:**
    - **The Happy Path:** The ideal route you designed.
    - **The Reality Path:** The most common path users actually took.
    - **Success Variance:** AI identifies if users found "shortcuts" or "workarounds" to reach the goal.
    - *Example:* "Users are ignoring the 'Menu' and using the 'Search Bar' to find the Product Page 80% of the time."
- **First-Click & Misclick Logic:**
    - **First-Click Success:** Statistically, if a user's first click is correct, they are 80% more likely to succeed. We track this as a primary KPI.
    - **Misclick Rate per Screen:** pinpointing exactly which screen in your Figma flow is the "Problem Child."

C. AI Proactive Analyst

The AI automatically generates **"Insight Cards"** that pop up as the data comes in. The AI does the work; the designer just reviews.

- **Automated Pattern Detection:** The AI identifies recurring issues automatically.
    - *Example Card:* "System Alert: 8 out of 10 users are failing Task 2. Most users are clicking on the 'Logo' instead of the 'Menu' to find the login page."
- **Anomaly Detection:** AI flags when a result is weird or unexpected.
    - *Example Card:* "Anomaly Detected: Mobile users are finishing 50% faster than Desktop users. Check if the Mobile layout is hiding unnecessary steps."
- **Proactive Summaries:** Every task block has a 1-sentence AI summary of the "Real Story" behind the numbers.

D. AI Definition of Done (DOD)

The AI monitors the data to tell you when it’s safe to stop the test.

- **Statistical Significance Tracker:** AI tells you when the numbers are "real" and not just a coincidence.
    - *Example:* "Confidence Level: 95%. You have enough data to prove that users hate the new checkout button."
- **Saturation Alert:** AI tells you when you aren't learning anything new.
    - *Example:* "Saturation reached. The last 5 participants have made the exact same mistakes as the first 10. You can stop the test now to save credits."

3.11. AI Insight Report

The report is not just a dashboard; it is a **Structured Narrative** designed to be handed directly to stakeholders (Managers, Clients, Developers). It explains "The Why" and "The How to Fix" using the results as proof.

A. Executive Summary (The Story)

A high-level overview for people who don't have time to look at every chart.

- **The Verdict:** A simple "Yes/No" on if the design is ready for development.
- **KPI Impact:** How this design will affect the business.
- **Example:** *"Verdict: Not Ready. Users love the visual style, but the checkout flow is too confusing, which will lead to a high cart abandonment rate."*

B. Issue Cards (The Evidence)

Detailed cards for every major problem found, sorted by how "bad" they are.

1. **The Issue:** Simple name (e.g., "Invisible Search Bar").
2. **The Impact:** Who was affected? (e.g., "75% of users couldn't find the search function").
3. **The Evidence:** A direct screenshot of the Confusion Heatmap or a quote from a user.
4. **The Recommendation:** Exactly what to change in the design.
    - *Example:* "Issue: Pricing is unclear. Evidence: Confusion heatmap shows 50 rage clicks on the 'Pro' plan header. Recommendation: Make the 'Contact Us' button more visible and add a price list."

C. Performance Comparison (A/B Testing)

If you are testing two designs, the AI compares them head-to-head.

- **Winner Selection:** AI picks the winner based on success rate and friction.
- **Example:** *"Design B is the winner. Users finished 20% faster and reported 15% less confusion than Design A."*

D. Methodological Credibility (Advanced Numbers)

Proves the research is scientific.

- **P-SUS (Predicted System Usability Scale):** AI predicts how "usable" your app is on a scale of 0-100.
- **Cognitive Load Benchmarking:** Estimates how much "brain power" users had to use.
    - *Example:* "Task 3 has high cognitive load. Users spent 15 seconds looking at the screen before their first click."

E. Case Study Example (Hypothetical: Food Delivery App)

- **Background:** Testing the new "Checkout" flow for a food app.
- **Methodology:** Unmoderated Usability Test, 15 participants.
- **AI Executive Summary:** "The app is visually appealing, but the 'Add Coupon' step is a major hurdle. 60% of users failed because they couldn't find where to enter the code."
- **Detailed Finding:** "Block 2 (Address Entry): High Friction. AI detected 12 'Hesitation Loops' on the Zip Code field. Users weren't sure if it was required."
- **Evidence:** Confusion Heatmap shows users clicking the 'Back' arrow repeatedly when the Zip Code error appeared.
- **Recommendation:** "Move the Zip Code field above the City field and make the error message red for better visibility."

4. MVP Credit limit reached Conditions (credit limit)

4.1. Home/Dashboard

- When the AI credit limit is reached, users will no longer be able to generate studies using AI.
- Showing plan details when the user clicks the “Generate” button.

4.2. Canvas:

- When the AI credit (AI-generated canvas, voice-based moderated testing) limit is reached, AI-generated prompts will no longer be available. The user can still proceed by manually entering the study details.
- user masih bisa mengenerate canvas atau testing selama credit “free trials” masih ada. kalau credit “free trials” sudah habis maka user akan di prompt untuk melakukan upgrade plan.
- AI credit limit reached di voice pas preview sm testing

4.3. Result:

- When the credit limit is reached, any active testing sessions will be stopped. If a study has already been launched and the link has been shared, data collection will pause immediately, even if participants are in the middle of a session.

5. Product Information Architecture & Pages (MVP)

5.1 Home/Dashboard (AI-First Entry Point)

Serve as the primary entry point for creating new studies using AI, while allowing existing users to quickly continue recent work.

- **Page Structure:**

- **AI Study Generation Section:**
  - Research goal input
  - Research type selector (e.g. usability testing)
  - Enhance prompt action
  - Generate study action
- **Templates Section:**
  - Pre-built Userlenz templates
  - Entry point to view all templates
- **Recent Studies Section (Existing Users) (Mark - nice to have)** :
  - List of recently accessed studies

- **User Actions:**
  - Enter a research goal
  - Enhance the prompt using AI
  - Generate a new study using AI
  - Start a study from a template
  - Open a recent study

- **System Behavior:**
  - AI-generated study creation consumes AI credits
  - If AI credits are exhausted, the AI-generated creation is disabled
  - Users can still create a canvas question manually or from templates when AI credits are unavailable

5.2 Project Page

Serve as the primary entry point for creating new studies using AI, while allowing existing users to quickly continue recent work.

- **Page Structure:**
  - Project list view
  - Project tab: (Active & Archive)
    - Active project actions:
      - Archive
      - Rename
    - Archive project actions:
      - Restore
      - Delete permanently (if possible - **(Mark - nice to have)**)
  - Search and filters

- **User Actions:**
  - Create & rename
  - Archive or restore projects

- **System Behavior:**
    - Projects contain multiple studies
    - Project permissions apply to all studies within the project
    - Archived projects are hidden but retrievable
    - Deleted projects permanently on archive remove all associated studies and data

5.3 Study List Page

Serve as the primary entry point for creating new studies using AI, while allowing existing users to quickly continue recent work.

- **Page Structure:**
    - Study list view (Table View)
        - Study name
        - Study status (draft, live pause)
        - Integration (Figma or live website)
        - Participants
        - Updated on
        - Study action menu (Rename, archive, move study)
    - Search and filters

- **User Actions:**
    - Create & rename
    - Archive
    - Move the study to another project

- **System Behavior:**
    - The study page contains a list of studies
    - Archived studies are hidden but retrievable
    - Deleted studies permanently on archive remove all associated data

5.4 Study Template Library

Enable users to quickly create studies using predefined and best-practice templates.

- **Page Structure:**
    - Template categories (prototype or live web)
    - Template cards with description
    - Template preview
    - Create a study from a template action

- **Available Template:**

- **Usability Testing:**
  - Usability testing a new product
  - Run an early prototype test
  - Feature discoverability template
  - Collect insights on features
  - Test feature usability
  - Test mobile app usability
  - Validate live website sign-up funnel flow
  - Test live website navigation
  - Test your website sign-up flow
  - Run a Single Ease Question (SEQ) test
  - System Usability Scale (SUS) Template

- **User Actions:**
  - Browse templates
  - Preview template structure
  - Create a study from a template
  - Customize the template after creation

- **System Behavior:**
  - Templates are editable after creation

5.5 Study Detail Page (Canvas)

After a study is generated via AI Chat and assigned to a project, the user enters a Study Workspace consisting of three primary pages:

1. Canvas
2. Results
3. Report

These pages represent the full lifecycle of a study: Design → Observe → Synthesize.

5.5.1 Canvas Page (Edit Mode)

Allow users to review, refine, and prepare the AI-generated study flow before launching.

- **Page Structure:**
  - **Top Controls:**
    - Project folder name (read-only)
    - Study name (editable) → View Goal detail (goals can be regenerated only)
    - User indicator (avatar or initial placeholder)
    - **Preview study** button
    - **Launch** button (Open study control modal)
    - Study indicator (live or stopped)

- **Main Content:**
  - Visual Flow Builder (Canvas)
- **Side Panel:**
  - Block Editor panel (appears when a block is selected)
  - Comments panel (appears when the comment icon is selected)

- **User Actions:**
  - Rename the study name
  - Edit AI-generated blocks
  - Generate or enhance block content using AI
  - Add, duplicate, delete, or change block types
  - Preview the study like participant experience or pov
  - Launch the study

- **System Behavior:**
  - The canvas is pre-populated by AI based on the selected research goal
  - All changes are auto-saved
  - AI actions consume AI credits immediately
  - When AI credits reach the limit:
    - AI generation and enhancement actions are disabled
    - Manual editing remains available
  - Once the study is launched:
    - Block structure becomes locked
    - AI regeneration actions are disabled

5.5.2 Canvas Page (Preview Mode)
Allow users to experience the study exactly as participants will before launch.

- **Page Structure:**
  - Full-screen participant experience overlay
  - Welcome + Language Selection screen
  - Sequential study blocks
  - Audio instructions with transcript support
  - Preview navigation controls

- **User Actions:**
  - Start preview
  - Navigate through study blocks
  - Play audio instructions
  - Exit preview back to Canvas

- **System Behavior:**

- Preview reflects the latest saved Canvas version
- No participant data is recorded
- Preview does not consume AI credits
- Exiting preview returns the user to Edit Mode

5.5.3 Canvas Page – Live Study State (Launch & Control Modal)
Allow users to start, stop, and monitor an active study without leaving the Canvas Page, while preventing accidental edits during testing.

- **Page Structure:**
  - **Modal Overlay (on top of Canvas):**
    - Modal title: Study Control
    - Entry point: "**Launch**" button
  - **Study control content:**
    - Shareable participant testing link
    - Copy link action
    - Set participants' limit
    - **Primary Actions:**
      - "Start testing" (default)
      - Changes to "Stop testing" when the study is live
    - **Monitoring Section**
      - Total participants
      - Completed sessions
      - In-progress sessions
      - Drop-offs (if available)

- **Background Canvas State:**
  - All editing interactions are **disabled**
  - Visual dimming is applied to indicate the read-only state
  - Top controls (edit, regenerate, and add block) are **disabled**

- **User Actions:**
  - Start, stop, or resume testing
  - Copy participant link
  - Monitor live study progress
  - Close modal (return to Canvas in read-only, or move to result & report)

- **System Behavior:**
  - **When the modal opens:**
    - Canvas enters a **disabled state**
    - No edits are allowed
  - **When "Start testing" is clicked:**

        - Study status changes to Live
        - Participant data collection begins
    - **When “Stop testing” is clicked:**
        - Study status changes to Stopped
        - No new participants can enter the study
    - **When “Resume testing is clicked:** basically is the same behavior as starting testing
        - Study status changes to Live
        - Participant data collection begins again
    - **Closing the modal:**
        - Returns the user to Canvas
        - Canvas becomes editable only if the study is not live

5.5.4 Results Page View (Observe Mode)

The Results page is the real-time "Command Center" where designers observe participant behavior. Unlike traditional tools that just show numbers, Userlenz uses AI to proactively flag friction points and data quality issues as they happen.

Page Structure:

- **Top Controls (Consistent with Canvas):**
    - Tabs: [Canvas] [**Results**] [Report]
    - Status Badge: Live / Paused / Stopped
    - Participant Counter: (e.g., 12/20 Completed)
- **Left Sidebar (Summary Metrics):**
    - **Study Health Score:** AI-calculated score (0-100) based on data quality.
    - **Success Rate:** Aggregated percentage of task completion.
    - **Average Time:** Total session duration vs. intended duration.
    - **Misclick Rate:** Percentage of clicks on non-interactive areas.
- **Main Content (Data View Toggles):**
    - **Aggregated View:** Grouped data for all participants (Heatmaps, Funnels).
    - **Individual View:** List of every participant with their specific recordings and transcripts.
- **Proactive AI Sidebar (The Analyst):**
    - A feed of "Insight Cards" generated automatically by the AI (e.g., "Alert: 5 users struggled with Task 2").

User Actions:

- **Filter Data:** Segment results by device (Mobile/Desktop), Health Score, or Success/Failure.
- **Inspect Blocks:** Click on any specific block from the study flow to see its dedicated heatmap and response breakdown.
- **Review Individual Sessions:** Open a specific participant's log to view their

"Reality Path" and text answers.
- **Mark as Invalid:** Manually exclude a participant if the AI Health Score flags them as a "Speeder."

System Behavior:

- **Live Data Streaming:** Results update automatically without page refresh as participants finish.
- **AI Friction Detection:** The system automatically identifies "Rage Clicks" and "Hesitation Loops" and pins them to the block results.
- **DOD (Definition of Done) Monitoring:**
    - *Example:* Once 15 participants show the same error, the AI triggers a **Saturation Alert**: *"New participants are repeating known patterns. You have reached 95% insight saturation; additional testing may not yield new information."*
    - *Example:* Once the error margin is low, the AI triggers a **Statistical Significance Tracker**: *"Confidence Level: 95%. You have enough data to prove that the 'Checkout' button is causing 20% more errors than the original."*

5.5.5 Report Page View (Synthesize Mode)

The Report page is the final "Storytelling" layer. It automatically formats the results into a professional Case Study that can be handed directly to stakeholders. It focuses on "The Why" and "How to Fix."

Page Structure:

- **Top Controls:**
    - Tabs: [Canvas] [Results] [**Report**]
    - Export Button: (PDF / CSV / Share Link)
- **AI Executive Summary (The TL;DR):**
    - **The Verdict:** AI-generated status (e.g., "Ready for Dev" or "Needs Revision").
    - **KPI Impact:** Summary of how the design affects business goals.
- **Main Content (Case Study Layout):**
    - **Goal Recap:** Reminder of the original research objective.
    - **Issue Cards:** Structured cards showing found problems, sorted by severity (Critical, Major, Minor).
    - **The Evidence:** Embedded snapshots of "Confusion Maps" and key user quotes.
    - **The Recommendation:** Clear UI/UX fix suggested by AI.
- **Methodology Block:**
    - **P-SUS Score:** AI-predicted System Usability Scale (0-100).
    - **Cognitive Load Score:** AI estimate of the "Brain Power" required for the

tasks.

User Actions:

- **Edit Report:** Manually override AI recommendations or summaries before sharing. (Mark - **Optional/Nice to have**)
- **Add/Remove Evidence:** Choose which heatmaps or quotes appear in the final PDF. (Mark - **Optional/Nice to have**)
- **Generate Share Link:** Create a view-only public link for stakeholders.
- **Export to PDF:** Generate a high-fidelity document formatted as a design case study.

System Behavior:

- **Automated Pattern Clustering:** AI groups similar user complaints into a single "Issue Card" to avoid redundancy.
- **Cognitive Load Calculation:**
    - *Example:* If users hovered for 12 seconds before clicking, the system marks the block as **"High Cognitive Load"** and suggests simplifying the layout.
- **Evidence Linking:** Every recommendation is backed by a link to the specific data point in the Results page for transparency (The "Glass Box" principle).
- **Stakeholder Optimization:** The report defaults to a visual "Presentation Mode" layout that prioritizes clear headings and big visuals over raw data tables.

5.6 Credits / Plan Page

Manage subscription plans and AI credit usage.

- **Page Structure:**
    - Current plan overview
    - AI credit balance (AI Gen & AI Voice)
    - Available plans
    - Add on for each AI credit
- **User Actions:**
    - Purchase AI credits
    - Upgrade or downgrade plan
    - Cancel or reactivate subscription
- **System Behavior:**
    - AI credits are consumed only by AI-powered actions
    - When credits reach zero, AI features are disabled

    - The user can buy AI Credits

5.7 AI Credits & Plan Widget (Sidebar)

Provide real-time visibility into AI usage and plan status.

- **Displayed Information:**
    - Remaining AI credits (AI Gen & AI Voice)
    - Plan name
    - Plan renewal or expiration date
- **System Behavior:**
    - Updates in real time
    - Low-credit warnings are triggered
    - Clicking the widget opens the Credits / Plan page

5.8 Profile & Settings

Manage personal, workspace, and system preferences.

5.8.1 Profile Settings

Allow users to manage their personal identity and basic account information.

- **Page Structure:**
    - **Profile details:**
        - Full name
        - Email address
        - Profile photo/avatar
    - **Account Preferences:**
        - Default language
- **User Actions:**
    - Edit personal information
    - Upload or remove profile photo
    - Update account preferences language
- **System Behavior:**
    - Changes are saved automatically
    - Email changes require verification
    - Language changes apply immediately across the product

5.8.2 Notification Preferences

Give users control over how and when they receive system and collaboration updates.

- **Page Structure:**
  - **Email Notifications:**
    - Marketing email preferences
    - AI credits running low
    - Plan expiration reminders
  - **In-App Notifications:**
    - System alerts:
      - AI credits running low
      - Plan expiration reminders
    - Study updates (ex, participants reached the goal)

- **User Actions:**
  - Enable or disable notification types (in-app or email) → toggle
  - Mute specific notification categories

- **System Behavior:**
  - Preferences apply per user
  - Changes take effect immediately

5.8.3 Security & Privacy

Protect user accounts and provide transparency and control over data and access.

- **Page Structure:**
  - **Security:**
    - Change password
    - Two-factor authentication (2FA)
    - Active sessions
    - Connected devices
  - **Access Control**
    - Connected accounts (Google / SSO)
    - Connected app (Figma)

- **User Actions:**
  - Enable or disable 2FA
  - Log out of active sessions
  - Revoke connected devices or accounts

- **System Behavior:**

    - Security changes require re-authentication
    - Enabling 2FA enforces verification on the next login
    - Session revocation takes effect immediately
    - Privacy settings are enforced across analytics, communication, and data sharing

5.9 Help Center

The Help Center is a documentation-focused support area designed to help users find answers quickly through articles.

5.9.1 Help Center Overviews

Enable users to quickly find relevant help articles through search and category browsing.

- **Page Structure:**
    - **Search Bar**
    - **Popular / Recommended Articles**
    - **Article Categories:**
        - Getting Started
        - Study Creation
        - AI Features
        - Results & Analysis
        - Billing & Credits
        - Account
        - Troubleshooting
- **User Actions:**
    - Search for articles by keyword
    - Browse articles by category
    - Open an article
- **System Behavior:**
    - Search results are ranked by relevance
    - Categories are system-defined
    - **No direct support ticket** entry point is shown on this page

5.9.2 Article Detail Page

Provide detailed guidance to resolve specific questions or issues.

- **Page Structure:**
    - Article title
    - Last updated timestamp
    - Article content (text, images, links)

    - Related articles
    - Subtle Support Link (End of Article)

- **User Actions:**
    - Read the article content
    - Navigate to related articles
    - Click the "**Contact support**" link (optional)

- **System Behavior:**
    - Articles are maintained and versioned by the system
    - The "Contact support" option is only displayed at the end of the article
    - Support link is visually de-emphasized
    - Clicking the support link opens a simple support request form
    - Users may be prompted with related articles before submitting a ticket

6. Assumptions & Risks

- **Assumption:** Designers are willing to trust AI-generated suggestions and insights for their research.
- **Risk:** The quality of AI insights may not be consistently high, potentially leading to user mistrust. **Mitigation:** Always provide direct links to the source data (the video clip or quote) so users can verify the AI's findings.
- **Assumption:** The AI-generated voice may not be fully consistent throughout a study session, potentially affecting participant comprehension.
- **Assumption:** We can acquire or build a high-quality dataset to train our AI models for UX research best practices.
- **Risk:** The technical challenge of processing and analyzing video and audio data at scale could be significant. **Mitigation:** Begin with text-based analysis (surveys) for the MVP and iterate towards more complex data typ